

COMPUTER SCIENCES

CS 368-1 (2011 Summer) — Day 5 Homework

Due Tuesday, July 19, at the start of class.

Description

Write a Perl script to do unit conversions, focusing on the DRY principle and, thus, subroutines.

Details

The main point of this exercise is, of course, to use subroutines. But we want to write a useful script, too, so let's write a basic unit-conversion calculator. Your script should convert among a single type of units, such as length or mass or volume. **Note:** Your script must convert among **at least** three different units (e.g., meters, feet, and miles)! More is OK.

Here is one **possible** interaction style.

```
Convert from units: ft
Convert to units: m
Length in ft to convert: 42

42 ft = 12.8016 m

Convert from units: lb
Sorry, I don't know anything about 'lb' units!

Convert from units: q
Done!
```

Write the following three subroutines (at least):

- `get_valid_unit()`: Asks the user for a unit label (e.g., 'm' for meter, or 'ft' for feet) and accepts input. Checks the input against the valid unit labels. If the label is OK, returns the label; otherwise, prints a message and asks for the unit label again.
- `get_valid_number()`: Asks the user for a number and accepts input. Checks to make sure the input is an integer (see below for hints). If the number is OK, returns it; otherwise, prints a message and asks for a number again.
- `convert(number, from_unit, to_unit)`: Convert a number from one unit to another; returns the resulting number.

For input validation of numbers, it is sufficient to make sure each character is a digit. If you already know regular expressions, knock yourself out. Otherwise, you can step through each character in the input string and see if the character is greater than or

equal to the string '0' AND less than or equal to the string '9'. There are two bits of Perl that you may not know yet.

To get the length of a string, use the length function:

```
my $string_length = length($string);
```

To get the character at a certain position within a string, use the substr function (short for “substring”), documented [here](#).

```
my $character = substr($string, 4, 1); # fifth character of $string
```

Extra Challenge

Can you avoid defining the conversion factor between every pair of units? Imagine you have 10 different length units (inch, foot, mile, meter, etc.); there are $10 \times 9 = 90$ unique conversion pairs (e.g., inch $\rightarrow$ foot). But I claim you need only 10...

Reminders

Do the work yourself, consulting reasonable reference materials as needed; any reference material that gives you a complete or nearly complete solution to this problem or a similar one is not OK to use. Asking the instructors for help is OK, asking other students for help is not.

Hand In

A **printout** of your code on a **single sheet of paper**. Be sure to put your own name in the initial comment block of the code. Identifying your work is important, or you may not receive appropriate credit.